

Todo

September 6, 2026

Introduction • Motivate the problem

- Roadmap of thesis

Conclusion • What does it mean for it to be disproven?

Discussion • Open questions that remain?

- Is the disproof generalizable?

Prerequisites list • Cayley graphs

- Minimal cayley graphs
- No-lonely-colored graphs
- Girth
- Chromatic number
- Irreducible and Quasi-irreducible generating sets
- Hypergraphs
- Bridges in Hypergraphs
- R-uniform and Auxiliary hypergraphs
- Closed walks on hypergraphs
- Tranquil hypergraphs
- Gallai-Witt Theorem (can add proof (if not massive) if thesis too small)

Prerequisites writeup • Cycles in minimal cayley graphs can't have lonely colors proof or mention from paper

- example of no-lonely-colored graph that is not cayley?
- cayley graph example? => visualization of cayley graph
- distinction between irreducible vs quasi-irreducible example?
- auxiliary hypergraphs example?

Visualization of disproof technique • Potentially add coded animation?

Visualization of minimal Cayley Graph • Step by step edge coloring?

Writeup of Disproof •

Proper bibliography and references setup

Check latex against university standards

Cover Babai's 1995 revision of the conjecture • Find other attempts at proving or disproving the conjecture

Writeup of original conjecture with historical framing • Find out where motivation for conjecture came from

- Figure out proper timeline

Mindmap of concepts how they relate?